

Advanced Exploratory Data Analysis and Feature Engineering on an E-Commerce Transaction Dataset

Name

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Objective

The objective of this project was to clean and preprocess a raw e-commerce dataset and prepare it for machine learning and advanced analytics by applying statistical imputation, outlier treatment, exploratory data analysis, and feature engineering techniques.

Dataset Description

The dataset contains **1200 e-commerce transaction records** with information related to orders, customers, products, payment methods, coupons, and pricing details.

Important Columns

- OrderID
- Date
- Customer ID
- Product
- Quantity
- Unit Price
- Shipping Address
- Payment Method
- OrderStatus
- Tracking Number
- Items In Cart

- Coupon Code
- Referral Source
- Total Price

Methodology

1. Data Loading

The dataset was imported into a **Pandas Data Frame** using Python and inspected for structure, data types, and missing values.

2. Missing Value Handling

The Coupon Code column contained missing values. These values were replaced using **Mode Imputation**, which preserves the most frequently occurring category.

3. Outlier Detection and Treatment

Numerical columns (Quantity, Unit Price, Items In Cart, and Total Price) were analyzed using the **Interquartile Range (IQR)** method.

Extreme values were capped within the acceptable lower and upper bounds to reduce their impact on future machine learning models.

4. Feature Engineering

The following new features were created:

Feature	Description
Order Month	Month extracted from the order date
Order Year	Year extracted from the order date
Avg Item Value	Average value paid per purchased item
Cart Efficiency Ratio	Ratio of purchased items to items present in the cart

5. Exploratory Data Analysis

The following visualizations were generated:

- Distribution plot for Total Price

- Correlation heatmap for numerical variables
 - Bar chart showing orders by Payment Method
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Results

- Successfully handled missing values.
 - Treated outliers using the IQR method.
 - Created four predictive features.
 - Generated a cleaned and feature-enriched dataset suitable for machine learning and business analytics applications.
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Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - SciPy
 - Jupyter Notebook
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Conclusion

This project successfully transformed a raw e-commerce dataset into a clean, structured, and machine-learning-ready dataset. Statistical preprocessing, outlier treatment, feature engineering, and exploratory data analysis techniques were applied effectively. The resulting dataset is suitable for advanced analytics and predictive modeling tasks and demonstrates practical data science skills required in real-world industry projects.